

Value Leakage in Frequent Flyers Program Analysis

Prepared by *Irfan Ridzky Anto*



Data and methodology

1 **Data Source**

Airline loyalty program data from a fictitious airline. Source: Maven Analytics.

A canadian-based fictional airline that just held a campaign in 2018.

2 **Data Aggregation**

Aggregate and filter data to users who signed up in 2018 only to reduce tenure bias.

3 **Statistics Test**

A significance test is done to find out whether the number is statistically significant or no.

4 **The Flyer Points Leakage**

Median is used to compare financial leakage. Data between CLV - Dollar cost points redeemed. When the result is negative, meaning there is a leakage.

The 2018 Promotion successfully acquired highly engaged, profitable travelers with zero financial leakage

16.6% Churn Rate

Promo users exhibited a lower ‘Real Churn’ rate than standard enrollments (19% churn)

6x more flights

Promo travelers flew around 6x more often with a CLV almost equal to standard travelers

0.0% Leakage

0.0% of promo travelers caused financial leakage. Their redemption costs were absorbed by CLV

While the 2018 promotion drove rapid acquisition, some challenges remained unresolved.

1 **Massive User Acquisition**

The campaign aggressively expanded the Frequent Flyer base.

2 **Masked Attrition**

Standard churn report may miss ‘Silent Churn’, travelers who just stop flying. So real churn is silent churn + explicit churn.

3 **The Discount Travelers**

Did the airline subsidize low-value travelers who artificially inflate engagement metrics?

4 **Financial Leakage**

Are we paying massive points redemption to users who don’t generate revenue?



We tested the 2018 cohort across three phases to uncover the campaign's true ROI.

#1 The Hidden Loss

How many travelers abandoned the airline when accounting for invisible drop-offs? (silent churn)

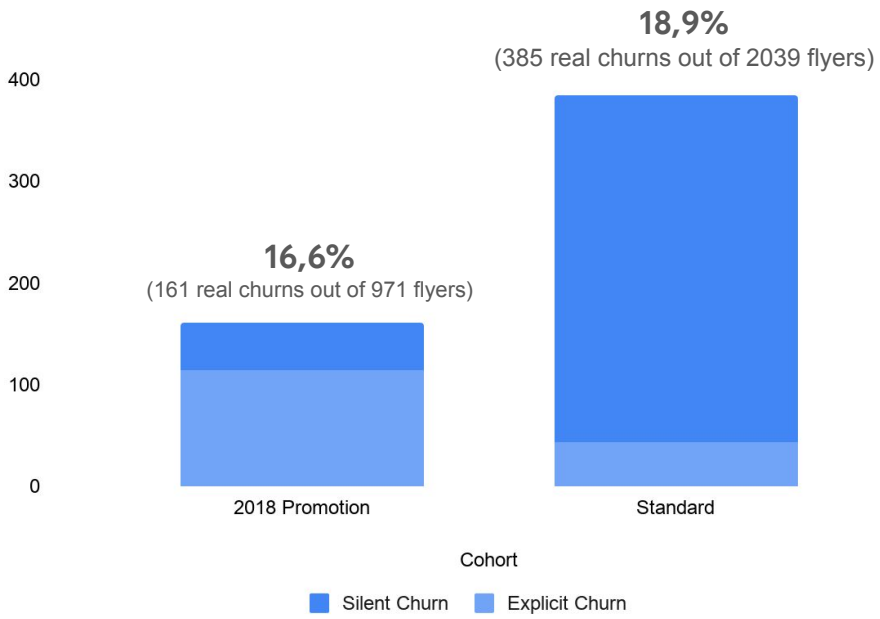
#2 Promo Quality

Did we acquire sustainable high-value flyers or transient discount hunters?

#3 Points Leakage

Did the dollar cost of points redeemed cannibalize the lifetime value of the customer?

#1 Contrary to conventional wisdom, users acquired via the 2018 Promotion exhibited lower real churn



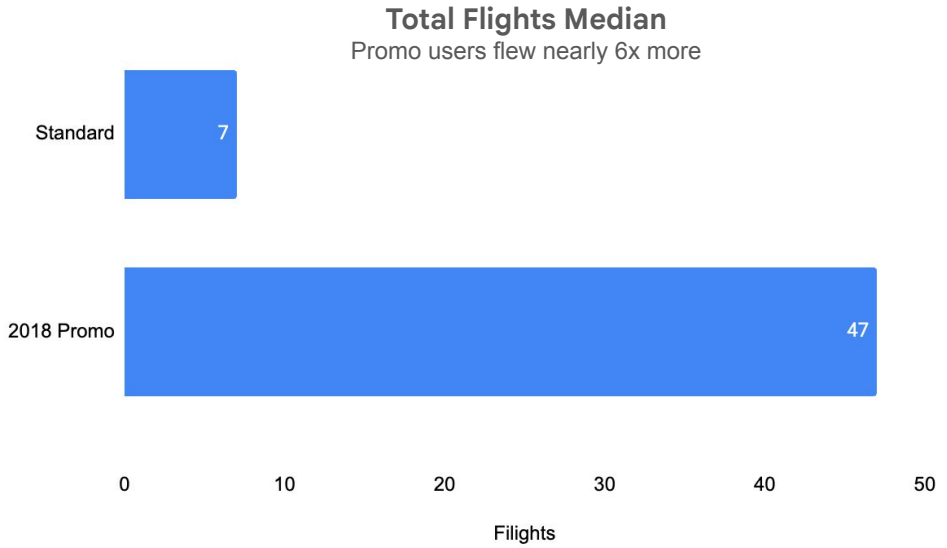
Key takeaway: The promotional campaign successfully acquired sticky travelers and retained them at around 2% higher than the standard enrollment

#2 Promo travelers demonstrated massive flight engagement while maintaining an on par CLV with standard users.

Statistical reality: Hypothesis testing confirms the CLV difference between groups is insignificant (p = 0.58)

The travelers value

The median CLV for promo travelers were **C$5,926** compared to **C$5,575** standard travelers



The promotion did not weaken travelers quality, despite showing neutral significance



#3 High dollar point redemption only triggers financial leakage if low CLV users drain subsidies without matching with revenue.

Customer Lifetime Value (CLV)
(The total revenue of travelers)

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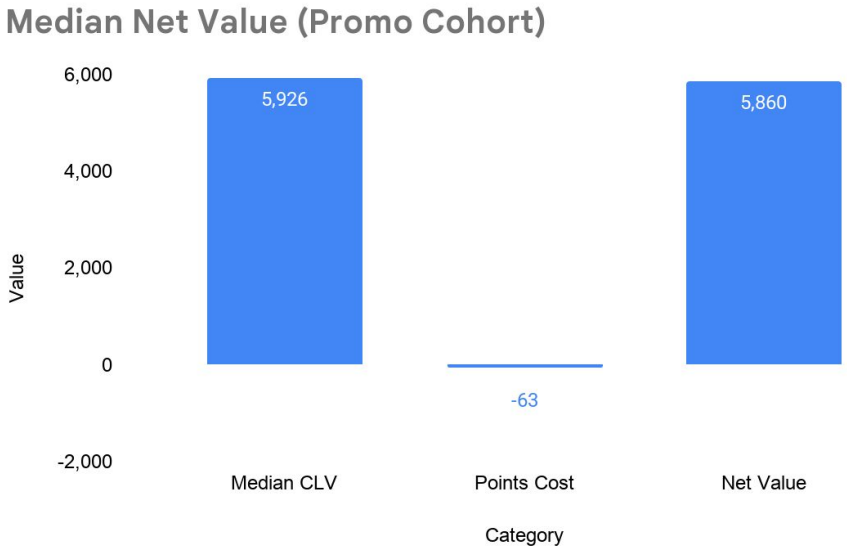
Dollar Cost of Points Redeemed
(The cost paid by the airline)

=

Net Value
(The profitability of user)

A leakage happens when Net Value <0.

#3 Although promo users redeemed more points, their inherent CLV ensured zero financial leakage for the airline.



Key takeaway: Out of 971 Frequent Flyers, no travelers caused leakage. High redemption did not cause leakage.

Across all phases, the 2018 promotion proved to be an effective strategy for profitable customer acquisition.

	Standard	2018 Promotion
Real Churn Rate	18,9%	16.6%
Median Flights/Yr	7	47
Median CLV	C$5,575	C$5,926
Leakage Rate	0.0%	0.0%

Key takeaway: Capitalize on the proven mechanics of the 2018 campaign to drive sustainable future growth

Update Reporting

Standardize the ‘Silent Churn’ (12+ month of inactivity) to ensure accurate calculation

Mimics The Success

The 2018 cohort’s success justify repeating this promotional structure.

Reframe Redemption

Shift focus from ‘reducing redemption cost’ to ‘maximizing engagement’. The data indicated high redemption correlates with profitable, engaged travelers.



Appendices

Google Colab Link
LinkedIn: Irfan Ridzky Anto